

Human Digestive System

Grade 6 STEM · Science Lesson

Learning Objectives

By the end of this lesson, students will be able to:

- Describe the basic process of digestion and why it matters.
- Identify the main organs involved in the digestive system.
- Explain what enzymes are and name three key digestive enzymes.
- Trace the journey of food from the mouth to the large intestine.
- Explain how nutrients are absorbed in the small intestine.

Subject: Science | Grade Level: 6 | Duration: 60 minutes

1. What Is Digestion?

Digestion is the process by which our body breaks down food into smaller nutrients that can be absorbed and used for energy, growth, and repair. It is a complex but fascinating journey that begins the moment food enters your mouth and ends when waste is expelled from the body.

Key Facts

- Digestion starts in the **mouth**.
- **Enzymes** are the chemical helpers that speed up digestion.
- Most **nutrient absorption** happens in the small intestine.
- Nutrients travel through the **bloodstream** to fuel every cell in your body.

2. Main Organs of the Digestive System

The digestive system is a complex network of organs that work together to process everything you eat and drink.

■ Mouth	Food enters and chewing begins. Amylase in saliva starts breaking down starch into sugars.
■ Oesophagus	Muscular tube that pushes food from the mouth to the stomach using wave-like contractions (peristalsis).
■ Stomach	Food is mixed with acids and pepsin. Proteins are broken down into amino acids.
■ Small Intestine	Nutrients are absorbed through finger-like villi into the bloodstream. Lipase from the pancreas digests fats here.
■ Large Intestine	Water is reabsorbed. Remaining waste is prepared for removal from the body.

■ Fun Fact!

Your stomach regenerates its own lining every **3 days** to protect itself from the powerful acids it produces — acids strong enough to dissolve metal!

3. What Are Enzymes?

Enzymes are biological catalysts — special proteins that speed up chemical reactions without being used up themselves. In digestion, they act like molecular scissors, cutting large food molecules into small, absorbable pieces. Without enzymes, digestion would be far too slow to keep us alive.

The Three Key Digestive Enzymes

Amylase

Where:

Saliva (mouth)

What it does:

Breaks starch into simple sugars.
You start digesting food while chewing!

Pepsin

Where:

Stomach

What it does:

Breaks proteins into amino acids.
Works best in the stomach's acidic environment.

Lipase

Where:

Pancreas

What it does:

Breaks fats into fatty acids.
Essential for nutrient absorption and energy.

How Enzymes Work — Step by Step

1. **Catalyst:** Enzymes speed up the breakdown of food molecules.
2. **Specificity:** Each enzyme has one specific job (e.g., amylase only breaks down starch).
3. **Lock & Key:** The enzyme's active site fits the food molecule like a key in a lock.
4. **Reusable:** After the reaction, the enzyme is released and can work again.

4. Nutrient Absorption in the Small Intestine

The small intestine is where most of the magic happens. After food has been broken down into tiny molecules by enzymes, the small intestine absorbs those nutrients into the bloodstream.

Feature	Detail
Length	Approximately 6 metres long — coiled inside the abdomen.
Inner lining	Covered in tiny finger-like projections called <i>villi</i> .
Villi	Massively increase the surface area available for nutrient absorption.
Absorption	Nutrients pass through the villi walls directly into the bloodstream .
Microvilli	Tiny hair-like structures on each villus further multiply surface area.

■ Did You Know?

If you could unfold and flatten all the villi and microvilli in your small intestine, the total surface area would be roughly the size of a **tennis court**!

5. Key Vocabulary

Digestion	The process of breaking down food into nutrients the body can use.
Enzyme	A protein that speeds up a chemical reaction — like a biological catalyst.
Amylase	An enzyme in saliva that breaks starch into simple sugars.
Pepsin	A stomach enzyme that breaks proteins into amino acids.
Lipase	A pancreatic enzyme that breaks fats into fatty acids.
Villi	Tiny finger-like projections lining the small intestine that absorb nutrients.
Absorption	The movement of nutrients from the small intestine into the bloodstream.
Peristalsis	Wave-like muscle contractions that push food along the digestive tract.
Nutrients	Substances in food (e.g. sugars, amino acids, fatty acids) that the body needs to function.
Catalyst	A substance that speeds up a chemical reaction without being consumed.

6. Test Your Knowledge

Answer the following questions to check your understanding. Discuss your answers with a partner before checking below.

Q1. Which enzyme breaks down starch, and where is it found?

Answer: Amylase — found in saliva, so starch digestion begins in the mouth.

Q2. Where does pepsin work, and what does it break down?

Answer: Pepsin works in the stomach. It breaks down proteins into smaller amino acids.

Q3. What is the role of lipase, and which organ produces it?

Answer: Lipase breaks fats into fatty acids. It is produced by the pancreas.

Q4. What part of the digestive system absorbs most nutrients?

Answer: The small intestine, thanks to millions of tiny villi lining its walls.

Q5. How do enzymes help digestion?

Answer: Enzymes act as catalysts — they speed up the chemical breakdown of large food molecules into smaller, absorbable nutrients.

Q6. How long is the small intestine, and why does its length matter?

Answer: About 6 metres. Its length provides a large surface area so the body has time to absorb all the nutrients.

7. Classroom Activity: Food's Amazing Journey

Activity: Map the Journey

Working in pairs, draw a simple diagram of the digestive system and label each organ. For each organ, write:

1. The name of the organ.
2. What happens to food there (mechanical or chemical digestion?).
3. Which enzyme (if any) is active at that stage.

Discussion Questions:

- Why do you think the small intestine is so long?
- What would happen if we had no enzymes?
- Why does the stomach need to regenerate its lining so often?

Summary

- Digestion breaks food into nutrients for energy and growth.
- Amylase, Pepsin and Lipase are the three key digestive enzymes.
- Food travels: Mouth → Oesophagus → Stomach → Small Intestine → Large Intestine.
- Nutrients are absorbed into the blood through the villi of the small intestine.
- Keeping your digestive system healthy means eating a balanced diet and staying hydrated!